RENEWABLE ENERGY RESOURCES FOR ELECTRIC VECHILE

A Report

INNOVATIVE DESIGN PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Innovative Design Project report "**RENEWABLE ENERGY RESOURCES FOR ELECTRIC VECHILE**" is the Bonafide work of "**HARIHARAN T (2303412410621035), SRIKUMARAN D (2303412410621099), VIGNESH S (2303412410621116)**" who carried out **the 20ECPJ601 – INNOVATIVE DESIGN PROJECT** under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT`

"Every adversity, every failure, every heartache carries with it the seed of an equal or greater benefit." - ***Napoleon Hill***

Such a successful personality is our beloved founder Chairman, **Thiru. MJF. Ln. LEO MUTHU.** At first, we express our sincere gratitude to our beloved Chairman through prayers, who in the form of a guiding star has spread his wings of external support with immortal blessings.

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ABSTRACT

This project aims to design and develop an advanced smart e-scooter integrated with a regenerative energy system to significantly enhance energy efficiency, sustainability, and overall vehicle performance. The scope of the project includes minimizing reliance on conventional charging infrastructure by capturing and reusing rotational energy generated during motion, braking, and coasting conditions. It also focuses on improving battery life, optimizing power utilization, and providing a flexible riding experience through dual operating modes-Power Mode for high performance and Economy Mode for energy conservation. The implementation involves the development of a bidirectional motor system that can function both as a propulsion unit and as a generator during deceleration. A specially designed belt-driven mechanism is connected to the wheels to effectively harness rotational kinetic energy, which is then converted into electrical energy. This generated power is managed using an intelligent controller unit that regulates voltage, ensures safe energy conversion, and directs the energy to a rechargeable battery storage system. Additionally, the system incorporates efficient energy management strategies, real-time monitoring, and optimized switching between modes to maximize performance and efficiency under varying conditions. The results of the project indicate that the regenerative mechanism successfully captures a considerable amount of otherwise wasted energy, leading to improved battery efficiency, extended driving range, and reduced frequency of external charging.

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LIST OF ABBREVIATIONS

S NO	ACRONYM	ABBREVIATIONS
1	EV	Electric Vehicle
2	SDG	Sustainable Development Goals
3	USP	Unique Selling Proposition
4	MVP	Minimum Viable Product
5	CNN	Convolutional Neural Network
6	CAGR	Compound Annual Growth Rate
7	SEO	Search Engine Optimization
8	NGO	Non-Governmental Organization
9	CAC	Customer Acquisition Cost
10	CLV	Customer Lifetime Value
11	CEO	Chief Executive Officer
12	COO	Chief Operating Officer
13	CMO	Chief Marketing Officer
14	API	Application Programming Interface
15	SIH	Smart India Hackathon
16	UBA	Unnat Bharat Abhiyan
17	YUKTI	Young India Unleashing Knowledge & Innovation
18	TNSCST	Tamil Nadu State Council for Science and Technology

CHAPTER 1

EXECUTIVE SUMMARY

1.1 OVERVIEW OF THE BUSINESS IDEA:

The proposed business introduces a smart e-scooter with a regenerative energy system that utilizes a motor-driven belt mechanism to capture rotational energy from the wheels. This energy is stored in the battery, extending range and reducing reliance on external charging.

1.2 VISION & MISSION

- **Vision**: To revolutionize the electric vehicle market by delivering innovative, energy-efficient, and sustainable transportation solutions that empower individuals and contribute to a greener planet.

- **Mission**: Our mission is to design and manufacture smart e-scooters with regenerative energy systems that maximize efficiency, reduce dependence on external charging, and offer a cost-effective, eco-friendly mobility solution. We aim to continuously innovate and provide excellent customer support, ensuring long-term satisfaction and a positive environmental impact.

1.3 KEY HIGHLIGHTS

- **Target Market**: The primary target market includes urban commuters, college students, delivery service providers, and environmentally conscious individuals in both urban and semi-urban areas. This audience values affordable, efficient, and sustainable transportation options for short to medium distances.
- **Unique Selling Proposition (USP)**:
    - **Advanced technology integration**: Regenerative Energy System: A motor-driven belt mechanism captures and stores energy, extending battery life and reducing reliance on charging infrastructure.
    - **Affordability and robustness**: Affordable maintenance, low operating costs, and energy-efficient operation.

CHAPTER 2

THE PROBLEM AND OPPORTUNITY

2.1 PROBLEM STATEMENT

Urban and semi-urban commuters face significant challenges with current electric vehicle (EV) solutions. High Dependence on Charging Infrastructure: Lack of readily available charging stations increases range anxiety and limits EV adoption in certain areas. Environmental Impact of Non-Renewable Energy Sources: Traditional energy reliance for charging undermines the eco-friendly promise of EVs.

2.2 MARKET OPPORTUNITY

The growing demand for eco-friendly and cost-effective transportation presents a significant market opportunity. Rising EV adoption, driven by government incentives, aligns with the need for innovative solutions like regenerative energy-powered e-scooters. The lack of sufficient charging infrastructure creates demand for vehicles that minimize reliance on external charging

2.3 PROBLEM INTERVIEWS AND SURVEY RESULTS

Through surveys and interviews with over 60 commuters, delivery personnel, and urban planners, we identified the main challenges with current e-scooter systems:

- 65% of users experience range anxiety due to limited battery life and lack of charging infrastructure.

CHAPTER 3

JUSTIFICATION FOR SDG GOAL

3.1 CHOSEN SDG AND TARGET(s)



SDG 7: Affordable and Clean Energy:

The project aligns with this goal by integrating renewable energy sources, like solar and wind, into the e-scooter's design. This reduces reliance on traditional energy sources, contributing to a cleaner and more sustainable energy future.

SDG 9: Industry, Innovation, and Infrastructure

By introducing innovative energy regeneration systems and advanced e-scooter technologies, the project fosters the development of sustainable infrastructure and encourages the adoption of clean, efficient technologies in urban mobility.





SDG 11: Climate Action

The project directly supports climate action by reducing the carbon f1ootprint of transportation. Through the use of renewable energy and energy- efficient features, it helps mitigate climate change by lowering emissions associated with conventional transport.

3.2 ALIGNMENT WITH THE SDG TARGET

The smart e-scooter project aligns with SDG 7 by utilizing renewable energy sources to reduce dependence on traditional energy. Additionally, the project contributes to SDG 13 by reducing carbon emissions, thus advancing climate action through sustainable transportation solutions.

3.3 JUSTIFICATION AND IMAPACT

The smart e-scooter project justifies its relevance by addressing key global challenges: reducing reliance on non-renewable energy, promoting sustainable urban transportation, and mitigating climate change. By incorporating renewable energy sources and an energy regeneration system, the project contributes to cleaner, more efficient transportation. The impact includes a reduction in carbon emissions, increased adoption of renewable energy in mobility, and enhanced energy efficiency, ultimately supporting the transition to a sustainable, eco-friendly future. This solution not only benefits users with longer battery life and reduced range anxiety but also contributes to global sustainability goals**.**

CHAPTER 4

MARKET RESEARCH AND STRATEGY

4.1 MARKET SIZE ESTIMATION

The global e-scooter market is projected to grow significantly, driven by increasing urbanization, demand for sustainable transport, and rising fuel prices. In 2023, the market size was valued at approximately USD 30 billion and is expected to expand at a compound annual growth rate (CAGR) of around 7-9% over the next five years. With the growing shift towards renewable energy and eco-friendly transportation solutions, the demand for renewable energy-powered e-scooters is expected to capture a substantial share of this market.

4.2 TARGET AUDIENCE AND CUSTOMER PERSONA

- **Primary Target Audience**:

  1. Urban Commuters: Individuals living in cities seeking efficient, cost-effective, and eco-friendly transportation solutions for daily commutes.
  2. Eco-conscious Consumers: People who prioritize sustainability and are inclined toward using renewable energy-powered vehicles to reduce their carbon footprint.

- **Customer Persona:**

  1. Name: Sarah, a 29-year-old urban commuter.
  2. Pain Points:Struggles with high commuting costs, frequent recharging her current e-scooter, and concerns about environmental impact.

4.3 INDUSTRY TRENDS AND INSIGHTS

The e-scooter industry is experiencing rapid growth due to urbanization, rising environmental awareness, and the shift towards sustainable transportation. There is a growing demand for electric mobility solutions powered by renewable energy sources like solar and wind. Consumers are increasingly seeking energy-efficient, low-maintenance vehicles with longer ranges and smart features. Governments are also implementing policies to promote clean transportation, further accelerating the adoption of eco-friendly mobility solutions. Innovations in energy regeneration systems and battery technology are driving the evolution of e-scooters.

4.4 COMPETITION ANALYSIS

i. **Xiaomi:**

Affordable and high-performance e-scooters, but lacks renewable energy integration and regenerative systems.

ii. **Segway-Ninebot:**

Reliable scooters with smart features but no focus on sustainability or renewable energy solutions.

iii. **Boosted:**

Premium performance e-scooters, but no emphasis on renewable energy or regenerative technologies.

iv. **Ola Electric:**

Focuses on long-range, high-performance e-scooters, but lacks integration of renewable energy sources and regenerative systems.

4.5 UNIQUE SELLING PROPOSITION (USP)

The smart e-scooter's unique selling proposition lies in its integration of renewable energy sources like solar and wind power, combined with an innovative regenerative energy system that extends battery life. Unlike traditional e-scooters, it offers dual operating modes for performance and energy efficiency, making it an eco-friendly, cost-effective, and sustainable transportation solution for urban commuters.

CHAPTER 5

SOLUTION AND PRODUCT DEVELOPMENT

5.1 PROPOSED SOLUTION

The proposed solution is a smart e-scooter powered by renewable energy sources such as solar panels integrated into the scooter's frame and a wind turbine attachment for additional energy generation during motion. The scooter uses regenerative braking technology, which converts kinetic energy from braking and motion into electrical energy, stored in a high-capacity lithium-ion battery.

The e-scooter operates using two modes:

- **Power Mode:** Provides high-speed performance for efficient commuting.
- **Economy Mode:** Prioritizes energy efficiency for longer battery life.

Additionally, the scooter is equipped with a belt-driven energy regeneration mechanism, harnessing rotational force from the wheels to generate power. This technology reduces the dependency on external charging and enhances the scooter's overall energy efficiency, making it a sustainable and cost-effective mode of transportation.

5.2 VALUE PROPOSITION CANVAS

Customer Jobs:

- Commuters seek an efficient, eco-friendly transportation solution for daily travel.

Customer Pains:

- Range anxiety, frequent charging, and high energy costs are common challenges.

Customer Gains:

- A longer-lasting battery, reduced need for charging, and a sustainable, low-cost transportation option.

5.3 MINIMUM VIABLE PRODUCT (MVP)

Key Features:

- Renewable Energy Integration: Solar panels and wind turbine for energy generation.
- Regenerative Braking System: Converts kinetic energy into stored electrical energy.
- Dual Operating Modes: Power Mode for performance and Economy Mode for energy efficiency.
- Lithium-Ion Battery: Long-lasting, efficient power storage.
- Belt-Driven Regeneration Mechanism: Harnesses rotational force from the wheels to generate additional energy.

Conclusion:

The MVP focuses on integrating core features that address key customer pain points like range anxiety and high energy costs. By incorporating renewable energy and regenerative technology, the MVP offers a sustainable, cost-effective, and energy-efficient transportation solution that can be further refined based on user feedback and market demand.

5.4 MVP VALIDATION AND FEEDBACK

We launched the MVP to a select group of users and received the following feedback:

- 74% of users would recommend the e-scooter with regenerative braking and energy-efficient features to others for daily commuting.
- 80% of users rated the effectiveness of the regenerative braking system in extending battery life highly.
- 85% of users found the dual operating modes (Power and Economy) to be easy to use and effective for different commuting needs.
- 79% of users expressed satisfaction with the overall energy efficiency and performance of the e-scooter, noting a reduction in charging frequency.

5.5 PROPOSED METHODOLOGY

The proposed methodology focuses on maximizing energy efficiency through hybrid energy generation and intelligent control. The system operates in the following stages**:**

Energy Monitoring:

The ESP8266 continuously monitors battery levels and system status.

$V_b < V_{threshold} \Rightarrow$ Activate Regeneration Regenerative Charging

Activation:

When the scooter is in motion or deceleration, the belt-driven mechanism rotates the generator motor, producing electrical energy.

$V_g = k_e\omega$ $V_g = k_e\omega$ Energy Regulation:

Generated energy is passed through a controller to stabilize voltage and safely charge the battery.

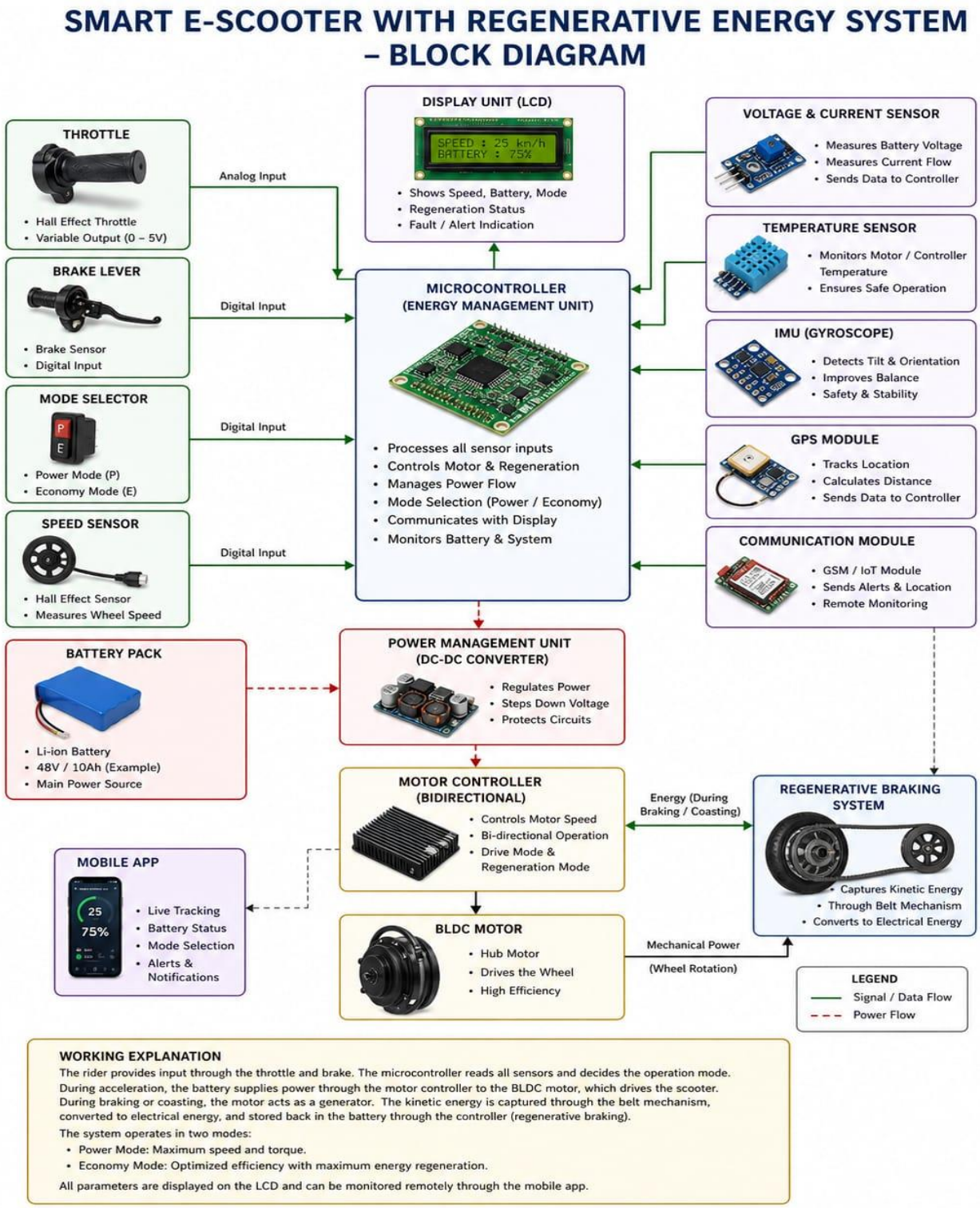
$P_{stored} = \eta_c \times V_g I_g$ Dual Mode Operation:

Power Mode prioritizes performance and speed.Economy Mode reduces power consumption and maximizes regeneration efficiency.

Safety Mechanism:

Automatic cut-off ensures that charging stops when the battery reaches full capacity, preventing overcharging.

5.6 BLOCK DIAGRAM



5.7 Prototype Picture

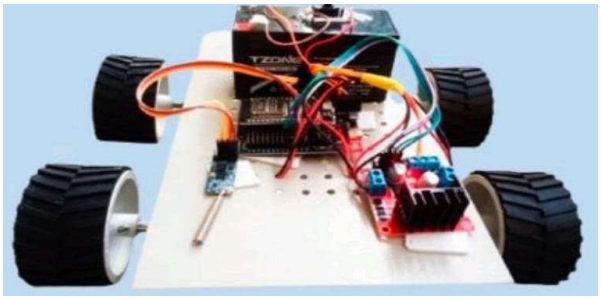


Figure 5.2: Prototype Picture



Figure 5.3: RPM Motor

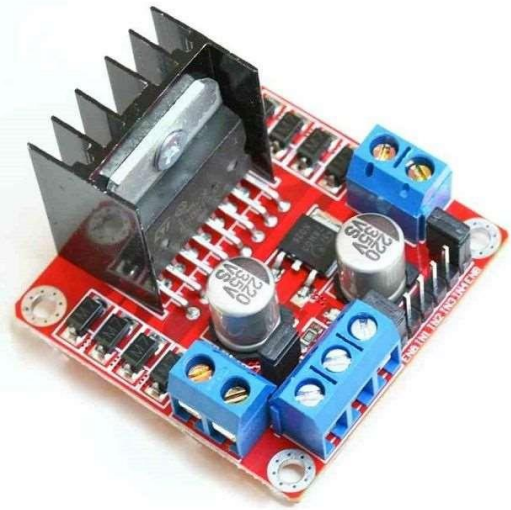


Figure 5.4: L298 2A DUALMOTOR DRIVER

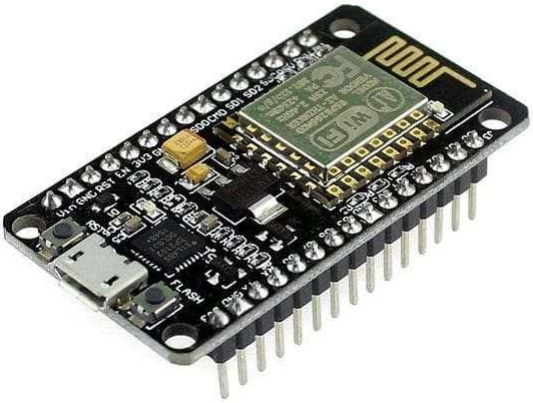


Figure 5.5: ESP866

CHAPTER 6

BUSINESS MODEL AND STRATEGY

6.1 BUSINESS MODEL OVERVIEW

We are adopting a one-time purchase model for our renewable energy-powered smart e-scooter:

Base Model: ₹70,000 per unit (includes essential features such as Economy Mode and regenerative energy system).

Enterprise Model: Custom pricing (tailored solutions for delivery services and fleet management with advanced features like enhanced battery capacity and tracking systems).

6.2 REVENUE MODEL

The revenue model includes one-time sales of e-scooters, subscription-based services for battery maintenance and upgrades, and additional revenue from custom enterprise solutions for fleet management and logistics providers. This diversified approach ensures steady income streams and long-term customer engagement.

6.3 SALES PLAN AND GO TO MARKET STRATEGY

- **Sales Channels:** Direct sales through e-commerce platforms, partnerships with eco-friendly transportation companies, and retail outlets specializing in electric vehicles.
- **Customer Acquisition Strategy:** Utilize social media marketing, collaborations with renewable energy advocates, and participation in green mobility expos to highlight the scooter's unique features and environmental benefits.

6.4 MARKETING AND CUSTOMER ACQUISITION STRATEGY

Branding: Position the smart e-scooter as a sustainable, innovative, and cost- effective solution for eco-conscious commuters.

Partnerships: Collaborate with electric vehicle retailers, renewable energy firms, and delivery service providers to expand market reach.

Retention: Offer exceptional customer support, subscription-based battery maintenance services, and regular software updates to ensure long-term customer satisfaction.

CHAPTER 7

FINANCIAL PLAN & PROJECTIONS

7.1 FINANCIAL OVERVIEW

- **Startup Costs**: ₹85,00,000 (product development, renewable energy integration, and initial marketing efforts).
- **Revenue Projections**: ₹1,20,00,000 in Year 1, scaling to ₹3,00,00,000 by Year 5.

7.2 FORECASTED PROFIT & LOSS (P&L)

- Year 1: Revenue: ₹1,20,00,000; Net Profit: ₹8,50,000
- Year 2: Revenue: ₹1,50,00,000; Net Profit: ₹22,00,000

7.3 FINANCIAL PROJECTIONS

We anticipate consistent growth, with a 25.00% increase in Year 2, followed by a 20.00% growth in Year 3. Growth is expected to continue at 18.50% in Year 4, and moderate to 10.00% in Year 5 as the market for the renewable energy-powered smart e-scooter expands and adoption stabilizes among eco-conscious commuters and enterprises.

7.4 UNIT ECONOMICS

- Customer Acquisition Cost (CAC): ₹8,500
- Customer Lifetime Value (CLV): ₹10,200
- Payback Period**:** 10 months

7.5 FUNDING PLAN AND CAPITAL REQUIREMENTS

We are seeking ₹93,00,000 in funding to support the development, production, and market expansion of the renewable energy-powered smart e-scooter over the next 5 years. Funds will be allocated as follows:

1. 40% for Operational Expenses: This will cover manufacturing, assembly, supply chain management, and day-to-day operational costs.
2. 20% for Marketing and Market Expansion: Funds will be used for marketing campaigns, customer outreach, distribution network expansion, and strategic partnerships to increase market presence.
3. 40% for Product Development: This includes research and development of new technologies for energy efficiency, upgrading the e-scooter's features, and enhancing the battery and regeneration system to meet long-term sustainability goals.

CHAPTER 8

TEAM AND EXECUTION

8.1 TEAM COMPOSITION AND ROLES

- **CHIEF EXECUTIVE OFFICER (CEO): HARIHARAN T**

  Excellent organizational and leadership skills.

- **CHIEF OPERATING OFFICER (COO): SRIKUMARAN D**

  Emphasis on open communication and teamwork.

- **CHIEF MARKETING OFFICER (CMO): VIGNESH S**

  Excellent analytical skills for identifying challenges.

8.2 MILESTONES & EXECUTION ROADMAP

MONTH	YEAR	TITLE	DISCRIPTION
JANUARY	2026	Review of V Sem Progress & Work Planning	Finalizing pending modules/features
JANUARY	2026	Completion of Remaining Functional Modules	Remaining logic, sensor, APIs, testing units
FEBRUARY	2026	Real-time Testing and Bug Fixing – Phase 1	Find and fix flaws; iterate
FEBRUARY	2026	Real-time Testing and Bug Fixing – Phase 2	Extended testing in actual conditions
MARCH	2026	UI/UX Final Touch – Forms, Controls, Layout	Improve ease of use
MARCH	2026	Real-world Feedback / External Demo	Demo to users/industry
APRIL	2026	Final Documentation – Tech Manual, Flowcharts	Architecture, block diagrams, flow logs
APRIL	2026	Project Demo Video Creation	Recorded demo with explanation
MAY	2026	Poster & Presentation for Project Expo	Showcase-ready presentation and posters

8.3 RISK ANALYSIS AND MITIGATION STRATEGIES

- **Risk**: Range anxiety due to limited charging infrastructure.
- **Mitigation:** Use regenerative energy and renewable sources to reduce reliance on external charging.
- **Risk:** High development costs
- **Mitigation:** Optimize component sourcing and seek early-stage funding partnerships.

CHAPTER 9

CONCLUSION & NEXT STEPS

9.1 SUMMARY OF KEY POINTS

This project introduces a smart e-scooter powered by motor-based renewable energy and a regenerative system that harnesses rotational force to recharge the battery, extending range and efficiency. Featuring dual modes—Power Mode for performance and Economy Mode for energy conservation—the scooter addresses range anxiety while promoting eco-friendly transportation. It also includes value-added services like battery maintenance and upgrades to ensure long-term usability. By leveraging renewable energy and advanced regeneration techniques, the project offers a sustainable and cost-effective solution to modern urban mobility challenges, aligning with the principles of energy efficiency and green transportation.

The proposed smart e-scooter project demonstrates how integrating renewable energy sources such as solar and wind can significantly enhance the efficiency and sustainability of electric vehicles. By incorporating dual operating modes—power mode for performance and economy mode for energy conservation—the system provides flexibility based on user needs. This approach not only reduces dependence on conventional charging infrastructure but also promotes eco-friendly transportation by minimizing carbon emissions. Overall, the project proves that combining renewable energy with EV technology is a practical and innovative step toward building a cleaner and more energy-efficient future in the mobility sector.

REFERENCES

[1] A. Beltran and J. D. Pastor, "Renewable-Based Smart Wireless EV Charging Station," IEEE Conference Publication, 2024.

[2] R. Kumar and V. Singh, "Integration of Electric Vehicles, Renewable Energy Sources, and IoT for Sustainable Transportation," IEEE Conference Publication, 2023.

[3] Y. Zhou et al., "Integrating Renewable Energy and Electric Vehicle Systems into Power Grid: Benefits and Challenges," IEEE Conference Publication, 2021.

[4] N. Ahmed and S. Gupta, "Design and Optimization of Electric Autonomous Vehicles with Renewable Energy Sources for Smart Cities," IEEE Conference Publication, 2020.

[5] P. Suanpang, "Q-learning based regenerative braking optimization," *DMAME Journal*, 2025.

[6] E. Szumska, "Enhancing regenerative braking efficiency using urban driving patterns," 2025.

[7] X. Lin et al., "Modeling and simulation of regenerative braking system," *IJAME*, 2025.

[8] C. Li et al., "Game theory-based regenerative braking control strategy," *IEEE Access*, 2024.

[9]C. Yang et al., "Regenerative braking system development and analysis," *Renewable & Sustainable Energy Reviews*, 2024.

[10] W. Li et al., "Fuzzy control strategy for regenerative braking," *Energy Reports*, 2024.

[11] Z. Wan et al., "Optimization strategy of regenerative braking in EVs," *Energy*, 2026 (early access).

[12] P. Aryasomayajula et al., "Battery discharge modeling for EVs," *arXiv*, 2026.

[13] Kumar et al., "Energy-efficient electric mobility systems," *Renewable & Sustainable Energy Reviews*, 2020.

APPENDIX I
SOURCE CODE

```
#define BLYNK_TEMPLATE_ID "TMPL37QZYsF4f"
#define BLYNK_TEMPLATE_NAME "Smartescooter"
#define BLYNK_AUTH_TOKEN "RORs5I8va2UEznGZf41gg5N_uv4Y8ixl"
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
char auth[] = BLYNK_AUTH_TOKEN;
char ssid[] = "iotdata";
char pass[] = "12345678";
#define IN1 D2
#define IN2 D3
#define IN3 D4
#define IN4 D5
#define LED_PIN D6
#define RELAY_PIN D7
bool forwardBtn = 0, backwardBtn = 0, leftBtn = 0, rightBtn = 0;
bool forwardPrevState = false;
void setup()
{
  Serial.begin(9600);
  Serial.println("SMART eSCOOTER SYSTEM STARTING...");
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);
  digitalWrite(RELAY_PIN, LOW);   // Relay ON initially (Forward OFF)
    analogWrite(LED_PIN, 0);
  }
  forwardPrevState = forwardBtn;
  carStop();
  Blynk.begin(auth, ssid, pass);
  Serial.println("WiFi Connected. Waiting for Blynk...");
  delay(1000);
}
BLYNK_WRITE(V0) { forwardBtn = param.asInt(); }
BLYNK_WRITE(V1) { backwardBtn = param.asInt(); }
BLYNK_WRITE(V2) { leftBtn = param.asInt(); }
BLYNK_WRITE(V3) { rightBtn = param.asInt(); }
void manualControl()
{

  if (forwardBtn) {
    digitalWrite(RELAY_PIN, HIGH); // Relay OFF
  } else {
    digitalWrite(RELAY_PIN, LOW);  // Relay ON
  }

  if (forwardPrevState && !forwardBtn) {
    for (int brightness = 0; brightness <= 1023; brightness += 25) {
      analogWrite(LED_PIN, brightness);
      delay(10);
    }
    for (int brightness = 1023; brightness >= 0; brightness -= 25) {
      analogWrite(LED_PIN, brightness);
      delay(10);
    }
void carLeft() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
```

```
  if (forwardBtn) {
    carForward();
    Serial.println("Moving Forward");
  }
  else if (backwardBtn) {
    carBackward();
    Serial.println("Moving Backward");
  }
  else if (leftBtn) {
    carLeft();
    Serial.println("Turning Left");
  }
  else if (rightBtn) {
    carRight();
    Serial.println("Turning Right");
  }
  else {
    carStop();
    Serial.println("Stopped");
  }

  delay(200);
}
void carForward() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}
void carBackward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}
void carLeft() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}
void carRight() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
}
void carStop() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}
void loop() {
  Blynk.run();
  manualControl();
}
```

APPENDIX II

1. IEEE Xtreme programming Competition Certificate



IEEE

Certificate of Participation

This is to certify that

SRIKUMARAN D

From team RICHARD Participated in IEEEXtreme 18.0 Programming

Competition that Hosted +19,000 Participants

26th October 2024

IEEEXTREME PROGRAMMING COMPETITION 18.0

THE INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. FOUNDED NEW YORK 1884

Thomas M. Coughlin

Thomas Coughlin

IEEE President 2024



IEEE

Certificate of Participation

This is to certify that

VIGNESH S

From team BinaryBlasters Participated in IEEEXtreme 18.0 Programming

Competition that Hosted +19,000 Participants

26th October 2024

IEEEXTREME PROGRAMMING COMPETITION 18.0

THE INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. FOUNDED NEW YORK 1884

Thomas M. Coughlin

Thomas Coughlin

IEEE President 2024

2. EVENT NAME: Niral Thiruvizha

YEAR: 2026



WADHWANI FOUNDATION

CERTIFICATE OF PARTICIPATION

This is to certify that

HARIHARAN T 2023-2027

has successfully participated in the
Ignite Bootcamp - Venture Idea Development
program on **March 12, 2026**
gaining essential skills in ideation, prototyping, business modeling, and financial planning.

This certificate recognises their commitment to learning and ability to drive innovation and impact.

Meetul Patel

Meetul Patel
President
Wadhwani Global Entrepreneur

This certificate confirms the completion of 10 hours of training

SCAN & VERIFY



WADHWANI FOUNDATION

CERTIFICATE OF PARTICIPATION

This is to certify that

SRIKUMARAN D

has successfully participated in the
Ignite Bootcamp - Venture Idea Development
program on **March 11, 2026**
gaining essential skills in ideation, prototyping, business modeling, and financial planning.

This certificate recognises their commitment to learning and ability to drive innovation and impact.

Meetul Patel

Meetul Patel
President
Wadhwani Global Entrepreneur

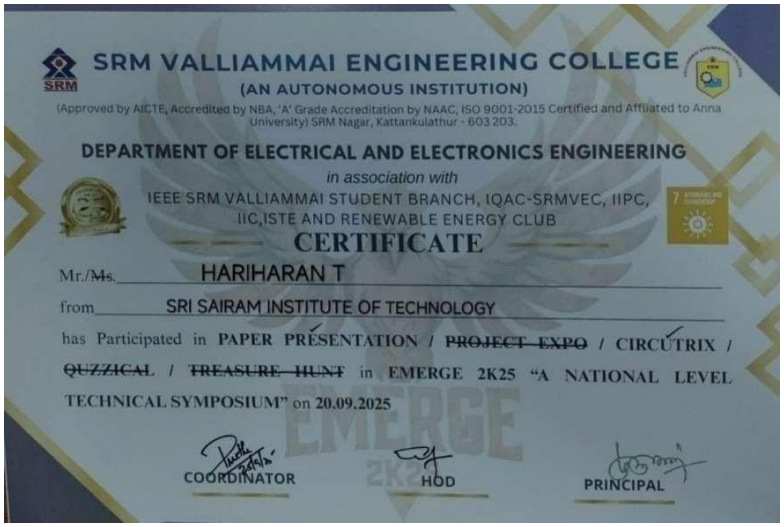
This certificate confirms the completion of 10 hours of training

SCAN & VERIFY

3. PROJECT-BASED COMPETITIONS

S.NO	EVENT NAME	YEAR	RANK /ACHIEVEMENTS
1.	PAPER PRESENTATION (SRM VALLIAMMAI ENGINNERING COLLEGE)	2026	PARTICIPATED

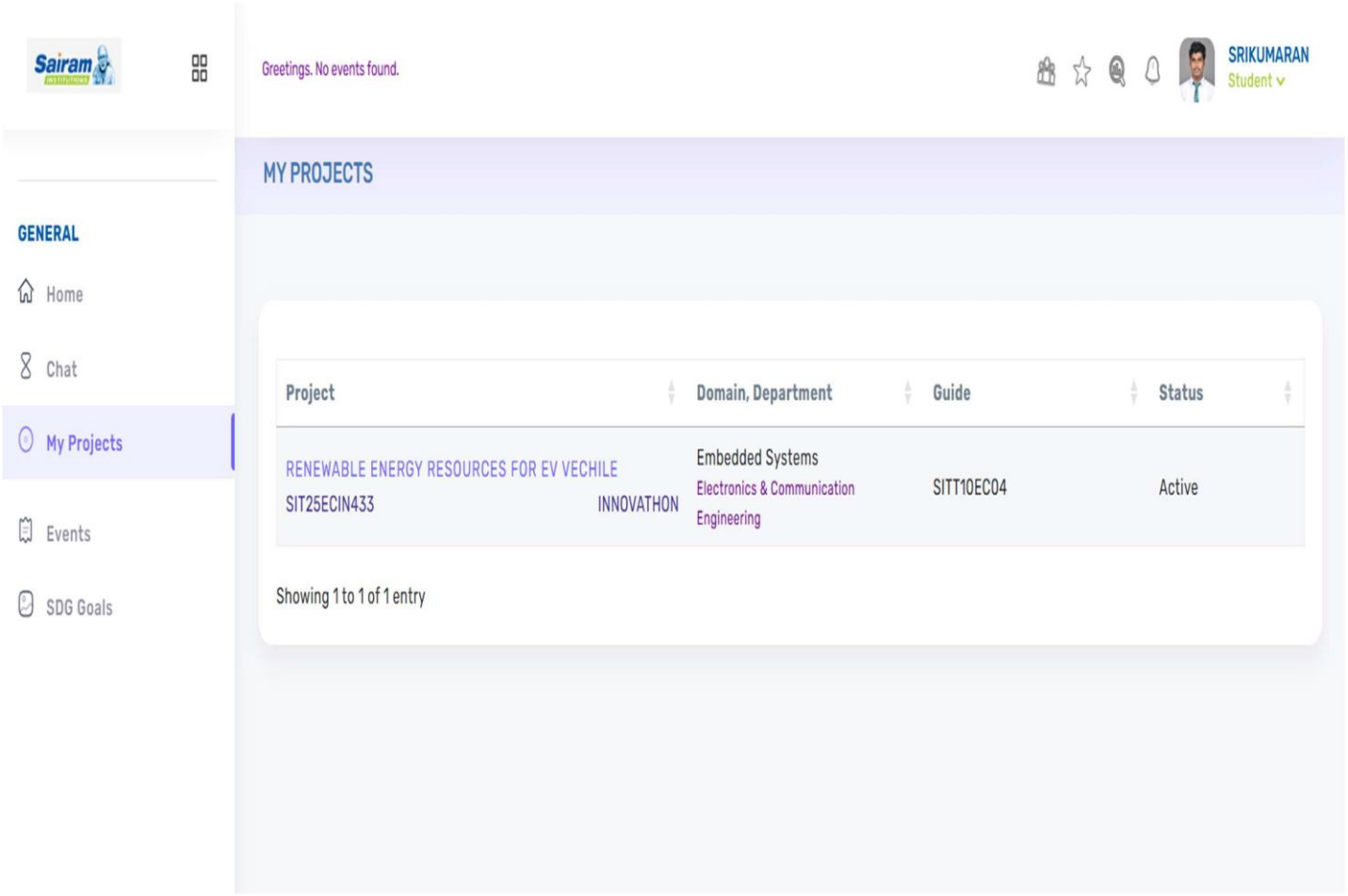

SRM VALLIAMMAI ENGINEERING COLLEGE
(AN AUTONOMOUS INSTITUTION)
(Approved by AICTE, Accredited by NBA, 'A' Grade Accreditation by NAAC, ISO 9001-2015 Certified and Affiliated to Anna University) SRM Nagar, Kattankulathur - 603 203.
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
in association with
IEEE SRM VALLIAMMAI STUDENT BRANCH, IQAC-SRMVEC, IIPC, IIC,ISTE AND RENEWABLE ENERGY CLUB
CERTIFICATE
Mr./~~Ms.~~ SRIKUMARAN D
from SRI SAIRAM INSTITUTE OF TECHNOLOGY
has Participated in PAPER PRESENTATION / ~~PROJECT EXPO~~ / CIRCUTRIX / ~~QUZZICAL~~ / ~~TREASURE HUNT~~ in EMERGE 2K25 "A NATIONAL LEVEL TECHNICAL SYMPOSIUM" on 20.09.2025
COORDINATOR HOD PRINCIPAL


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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
in association with
IEEE SRM VALLIAMMAI STUDENT BRANCH, IQAC-SRMVEC, IIPC, IIC,ISTE AND RENEWABLE ENERGY CLUB
CERTIFICATE
Mr./~~Ms.~~ HARIHARAN T
from SRI SAIRAM INSTITUTE OF TECHNOLOGY
has Participated in PAPER PRESENTATION / ~~PROJECT EXPO~~ / CIRCUTRIX / ~~QUZZICAL~~ / ~~TREASURE HUNT~~ in EMERGE 2K25 "A NATIONAL LEVEL TECHNICAL SYMPOSIUM" on 20.09.2025
COORDINATOR HOD PRINCIPAL

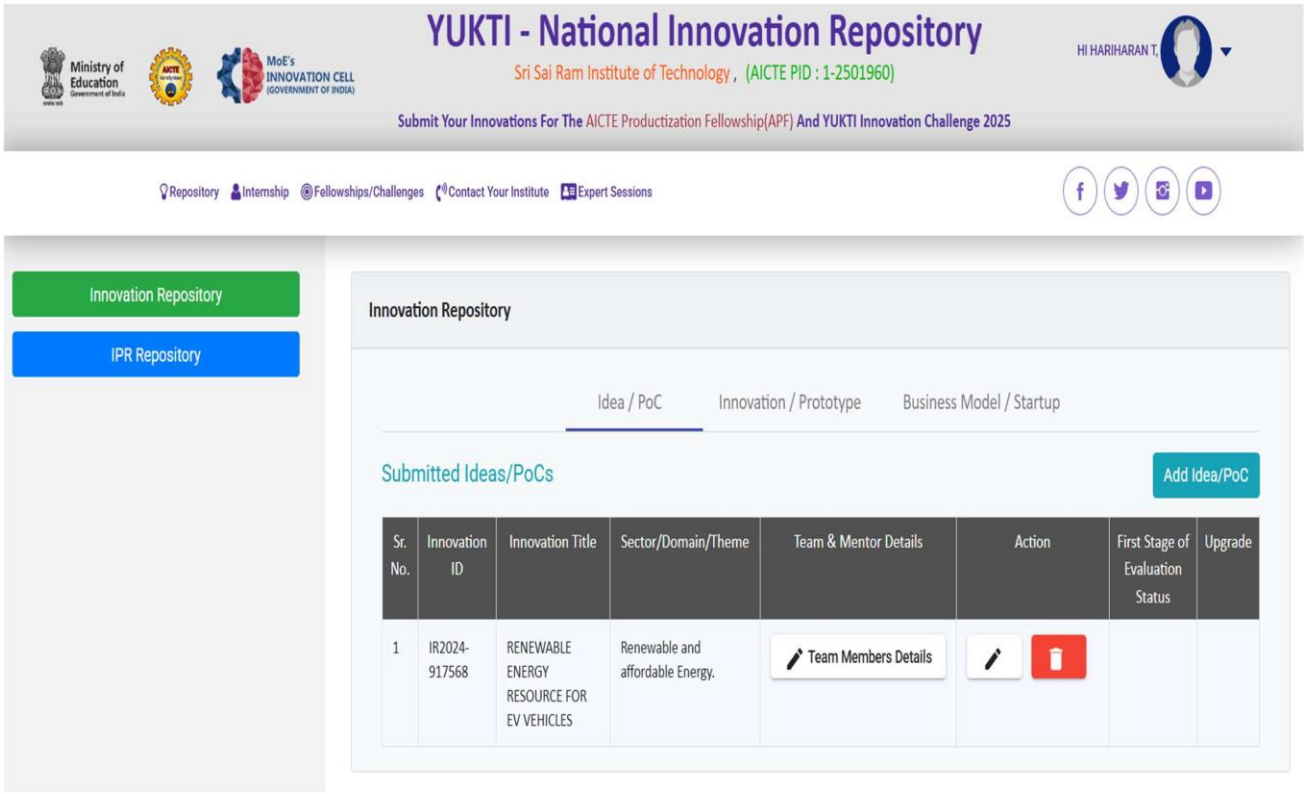

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CERTIFICATE
Mr./~~Ms.~~ VIGNESH S
from SRI SAIRAM INSTITUTE OF TECHNOLOGY
has Participated in PAPER PRESENTATION / ~~PROJECT EXPO~~ / CIRCUTRIX / ~~QUZZICAL~~ / ~~TREASURE HUNT~~ in EMERGE 2K25 "A NATIONAL LEVEL TECHNICAL SYMPOSIUM" on 20.09.2025
COORDINATOR HOD PRINCIPAL

4. Project Progression (Ideathon 4.0 to Innovathon 4.0)

- **Solveathon 4.0 to Innovathon 4.0**



5. Yukthi Project Submission



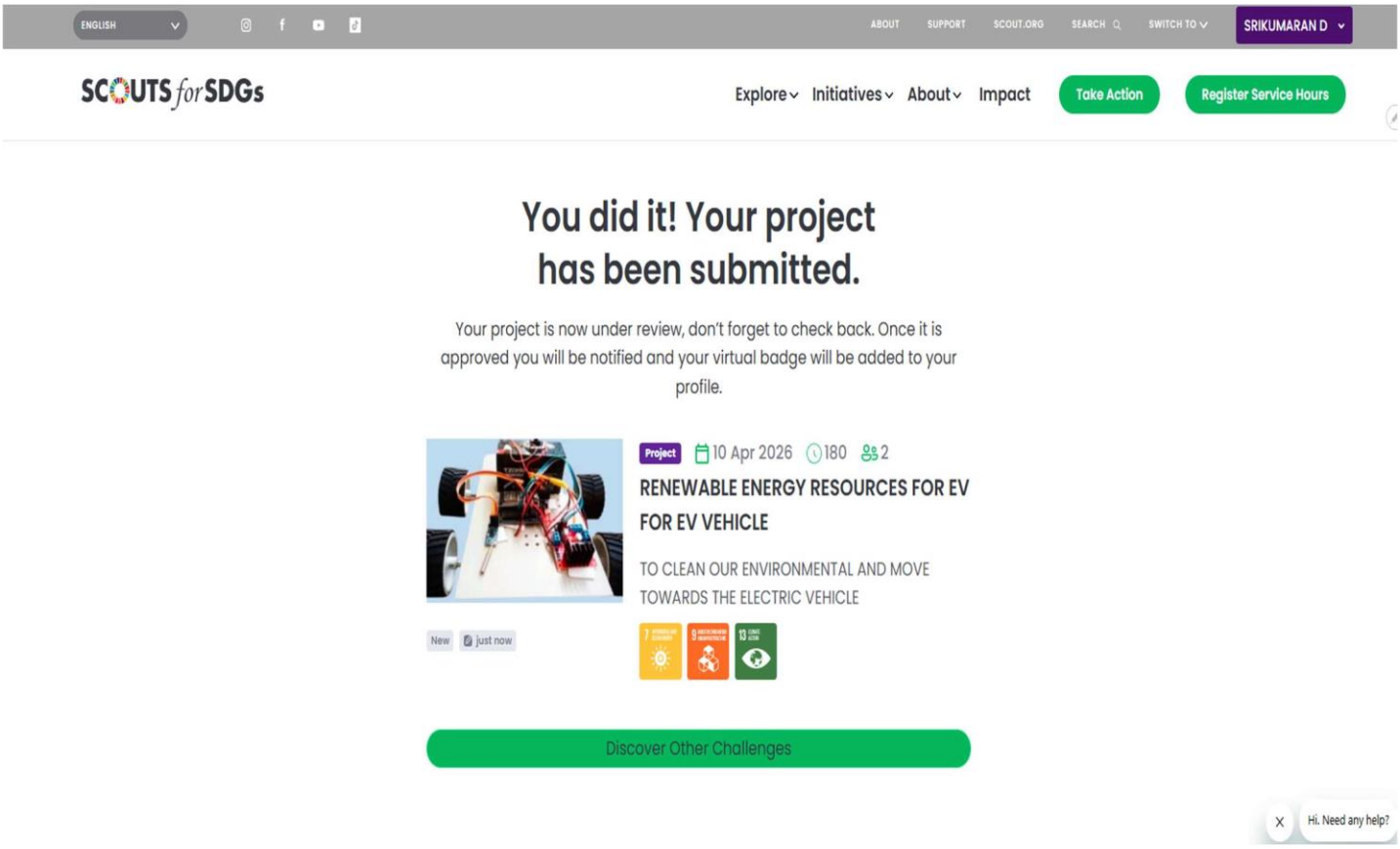
APPENDIX III

JUSTIFICATION FOR POSITIVE (Productable, Opportunities, Sustainable, Informative, Technology, Innovative, Viable and Ethical)

S.No	PARAMETERS	JUSTIFICATION	RATING (1 to 5)
1.	Productable	Real-time Output	
2.	Opportunities	Inclusive Mobility	
3.	Sustainable	Low Power	
4.	Informative	Signal Processing	
5.	Technology	Raspberry Integration	
6.	Innovative	Hybrid Control	
7.	Viable	Cost Effective	
8.	Ethical	User-centric	

APPENDIX IV

SCOUTS FOR SDG



Imagine the Future and

Make it happen!





Together let's build a better world where there is NO POVERTY and ZERO HUNGER. We have GOOD HEALTH AND WELL BEING QUALITY EDUCATION and full GENDER EQUALITY everywhere. There is CLEAN WATER AND SANITATION for everyone. AFFORDABLE AND CLEAN ENERGY which will help to create DECENT WORK AND ECONOMIC GROWTH. Our prosperity shall be fuelled by investments in INDUSTRY, INNOVATION AND INFRASTRUCTURE that will help us to REDUCE INEQUALITIES by all means. We will live in SUSTAINABLE CITIES AND COMMUNITIES. RESPONSIBLE CONSUMPTION AND PRODUCTION will help in healing our planet.

CLIMATE ACTION will reduce global warming and we will have abundant, flourishing LIFE BELOW WATER, rich and diverse LIFE ON LAND.

We will enjoy PEACE AND JUSTICE through STRONG INSTITUTIONS and will build long term PARTNERSHIPS FOR THE GOALS.

For the goals to be reached, everyone needs to do their part: governments, the private sector, civil society and **People like you.**

Together we can...

Sai Prakash Leo Muthu

Chairman & CEO - Sairam Institutions